OEL Task:

The list of practical is categorized into three areas namely:

1. Plant Location and Layout
2. Inventory Management
3. Project Management

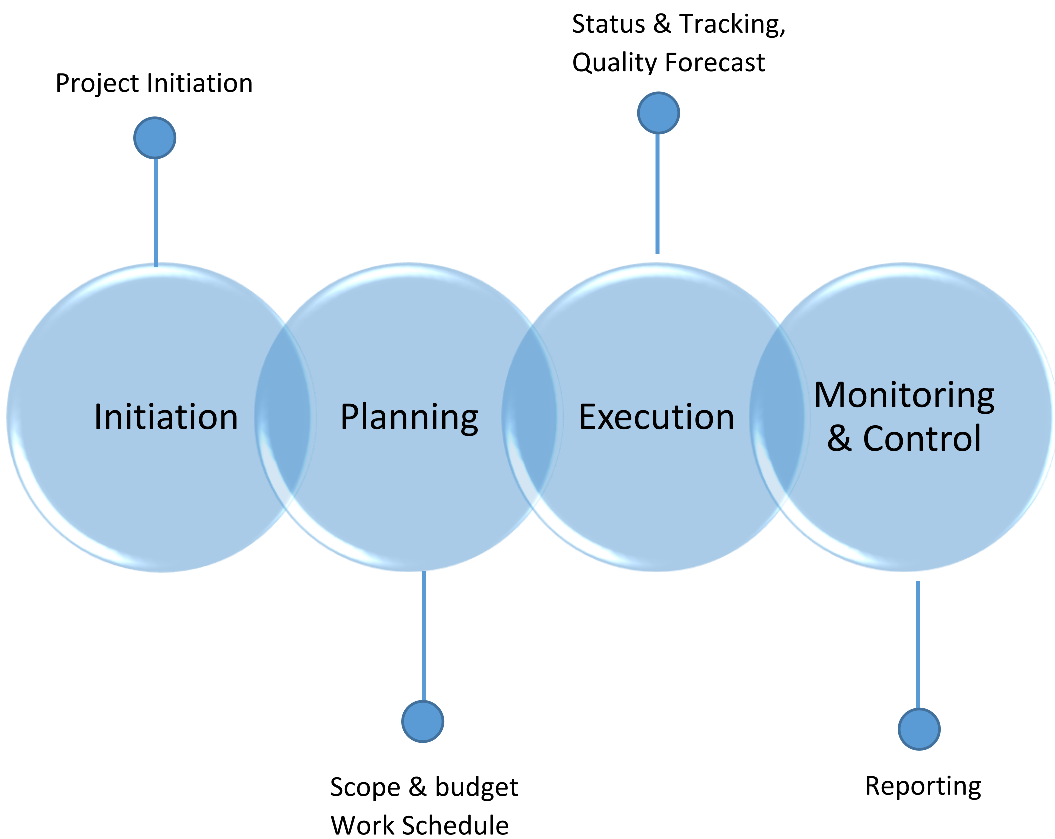
Develop a problem, in one of the above three areas, from an existing scenario of the industry or solve it on POM Software.

"Select a specific area project management related to the electric vehicle (EV) industry, focusing on challenges associated with the development of innovative technologies, unforeseen technical issues, project timeline management, and Assembly line"

Project Management:

Project Management involves planning and controlling a project, its goal to ensure that the project's objectives are archived on time and within budget.

Phases of Project Management



Project Scenario:

Developing an innovative solar tracker design in response to the increasing demand for efficient solar energy solutions presents several challenges. In our project management using POM-QM for windows challenges have arisen, including CPM-PERT like relying too much on one-time estimate, difficulty in using three time estimates, dilemmas in crashing activities, unexpected expenses straining our budget, and uncertainties in predicting project schedules. This report aims to thoroughly analyze these challenges and propose strategic solutions to enhance project management practices for improved accuracy and successful project execution within defined constraints.

1. Problem Description:

The primary problem revolves around the unexpected technical challenges in implementing the innovative solar tracker design with the agile eye mechanism. These challenges lead to delays, impacting time estimates and the project budget. The decision to crash activities introduces concerns about compromising the reliability of the solution. The uncertainties in standard deviation and variance calculations additional risks to the project schedule and assembly line.

Abstract:

This report presents an innovative solar tracker design featuring an agile eye mechanism aimed at optimizing energy. Emphasizing affordability and versatility, the mechatronics-focused project. In the realm of project management, challenges are highlighted, including reliance on cycle time estimates and unexpected expenses. The primary issue revolves around technical challenges causing delays and budget strain and assembly line. The report aims to analyze and propose solutions for improved project management practices.

2. Constraints:

- Unforeseen expenses are straining the project budget.
- Delays in the project due to technical challenges suggest a constraint on the project timeline.
- Issues arise from depending on one-time estimates and cycle time estimates and the need to crash activities in project management, adding constraints to planning and execution.
- Crashing activities raises concerns about compromising the reliability of the solar tracker design.

3. Assumptions:

- Assuming the effectiveness of the agile eye mechanism in optimizing energy.
- The assumption that the budget allocated for the project is sufficient to cover all anticipated costs.
- Assuming that one-time estimates and cycle time estimates are accurate and feasible in project planning.
- The assumption that the proposed strategic solutions will effectively address the identified challenges in project management.

4. Literature:

The literature review section provides an overview of existing research and relevant theories related to the solar tracker design, and project management.

- Explore this literature on solar tracker designs, especially those employing mechatronics and agile mechanisms.
- Review studies that focus on optimizing energy during cycle time and explore POM software by implementing the required operations activity in this process.
- Review literature on challenges faced in project management, particularly in the context of engineering and sustainable energy projects. Identify common issues and solutions proposed in the literature.
- Highlight key findings and technologies that have been employed in similar projects, Explore research on agile mechanisms in mechatronics for solar tracking systems.

5. Methodology:

The methodology section outlines the approach taken to conduct the project, including the design and project management aspects.

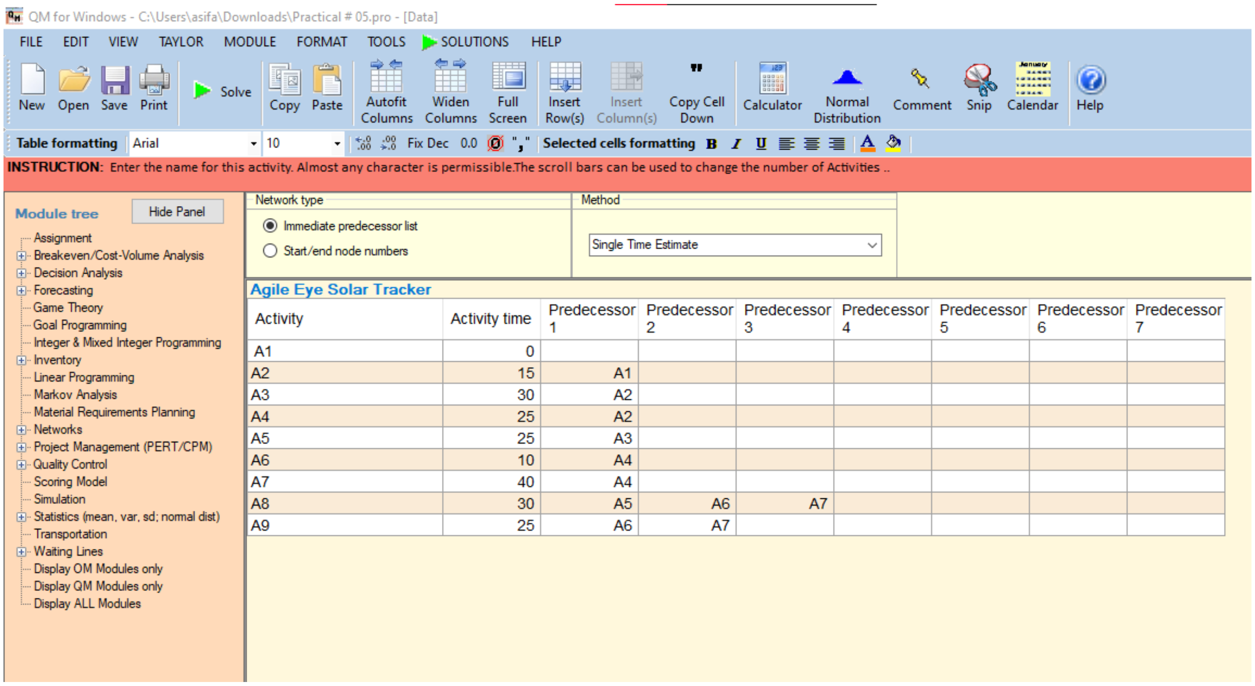
- Describe the process of conceptualizing, designing, and developing the solar tracker with the agile eye mechanism. Highlight any specific design principles, materials, or technologies used.
- Identify potential project management challenges, Implement risk assessment methodologies to identify and address potential issues
- Implement particular techniques for monitoring and controlling costs throughout the project.
- Use Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT) for time estimation and project scheduling. cost estimation methods for project budgeting.
- Implement Lean principles to optimize cycle time and minimize delays.
- project management tools, specifically POM-QM for Windows, were utilized.
- Describe the decision-making process, how estimates were generated, and how unexpected challenges were addressed.

The following activities are required during the operation of Agile eye solar tracker in the project management.

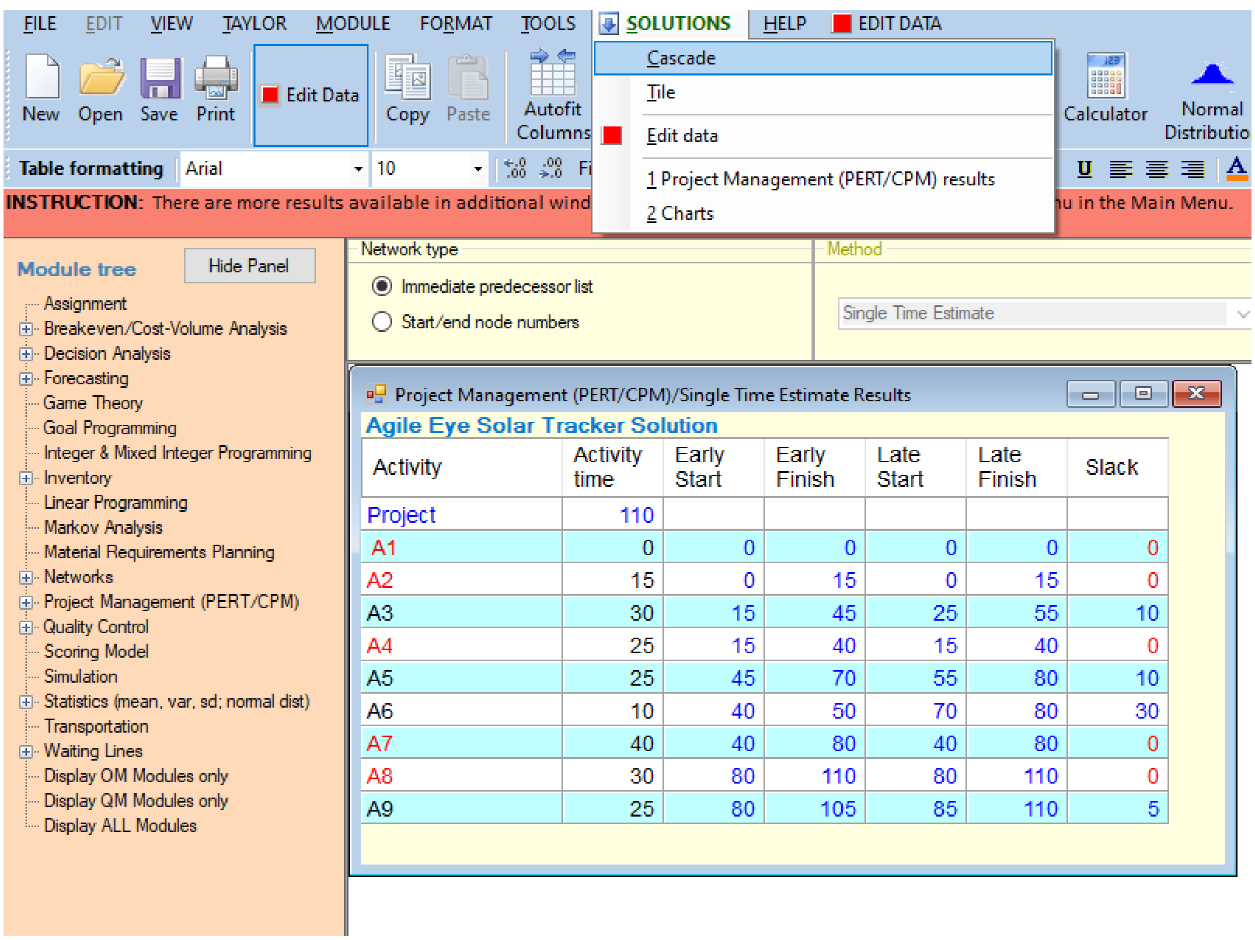
Serial No	Activity	Activity Code
01	**Agile Eye Mechanism**	**A1**
02	**Slidining System**	**A2**
03	**Base and Support**	**A3**
04	**Final CAD Designing**	**A4**
05	**Sensor Parsing**	**A5**
06	**Button Parsing**	**A6**
07	**Commanding Motors**	**A7**
08	**Solar Panels**	**A8**
09	**Complete Circuit**	**A9**

We are going to develop the precedence diagram for Agile Eye Solar Tracker on the basis of following activities to determine critical path; also find ES, EF, LS, LF and Slack for each activity. Using Critical Path Method (CPM).

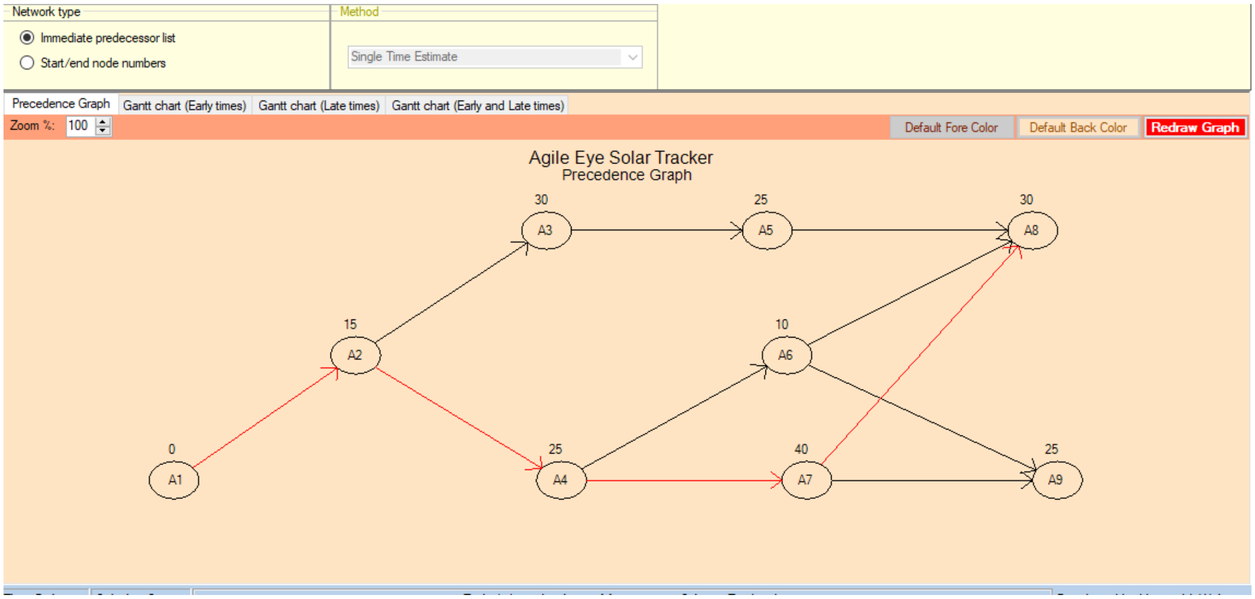
Activity	Immediate Procedure	Activity Time (days)
Agile Eye Mechanism A1	--------	0
Final CAD Design A2	Agile Eye Mechanism	10
Base and Support A3	Final CAD Design	26
Sliding System A4	Final CAD Design, Base and Support	14
Commanding Motor A5	Final CAD Design, Sliding System	9
Solar Panel A6	Final CAD Design, Sliding System	6
Complete Circuit A7	Final CAD Design, Commanding Motor, Solar Panel	15
Button Parsing A8	Final CAD Design, Commanding Motor, Complete Circuit	6
Sensor Parsing A9	Final CAD Design, Commanding Motor, Complete Circuit	14



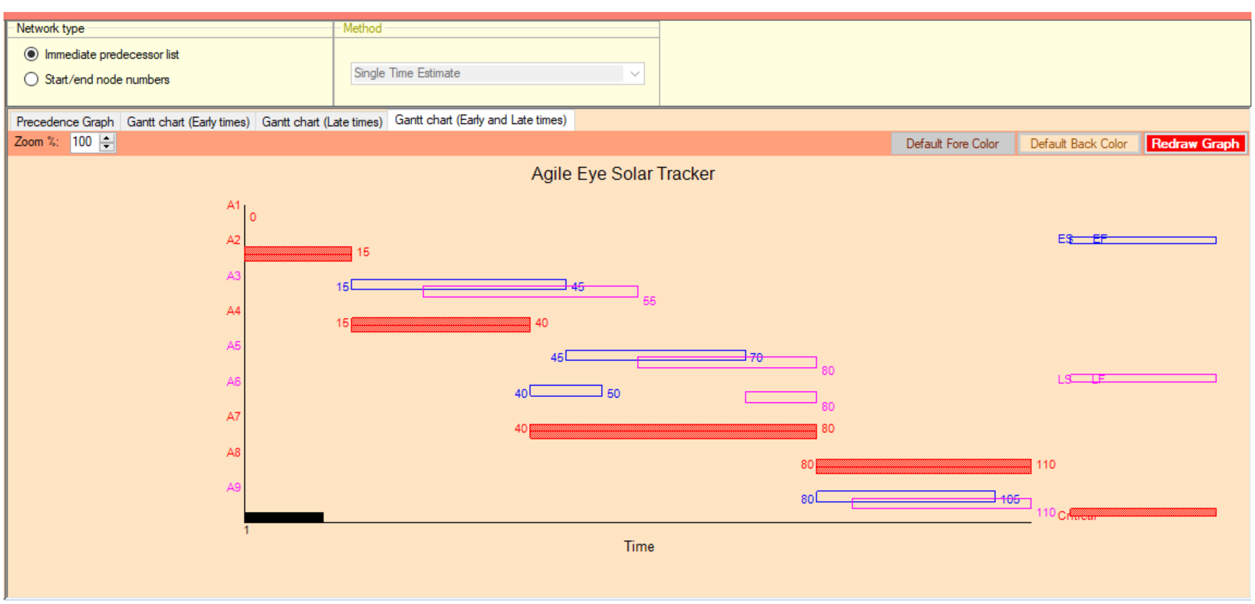
Activity	Activity time	Predecessor 1	Predecessor 2	Predecessor 3	Predecessor 4	Predecessor 5	Predecessor 6	Predecessor 7
A1	0							
A2	15	A1						
A3	30	A2						
A4	25	A2						
A5	25	A3						
A6	10	A4						
A7	40	A4						
A8	30	A5	A6	A7				
A9	25	A6	A7					



Activity	Activity time	Early Start	Early Finish	Late Start	Late Finish	Slack
Project	110					
A1	0	0	0	0	0	0
A2	15	0	15	0	15	0
A3	30	15	45	25	55	10
A4	25	15	40	15	40	0
A5	25	45	70	55	80	10
A6	10	40	50	70	80	30
A7	40	40	80	40	80	0
A8	30	80	110	80	110	0
A9	25	80	105	85	110	5



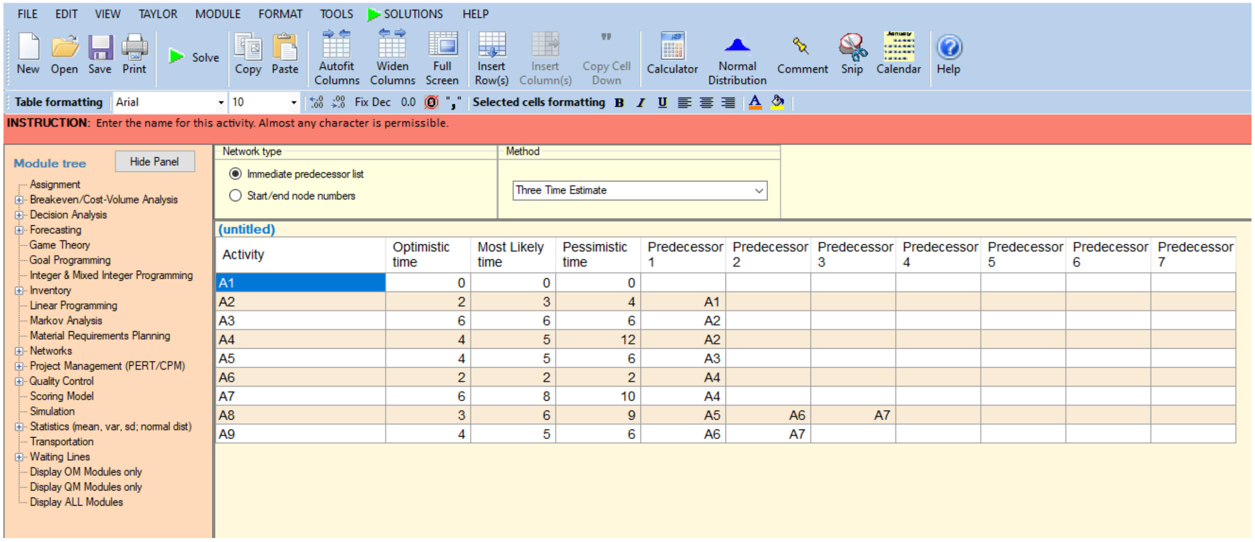
Interpretation: In the Critical Path Method (CPM) of project management, a network diagram, represented a precedence graph, is used to schedule and manage project tasks. Early Start and Early Finish are determined based on the logical relationships and durations of tasks, ensuring that the project can be completed as quickly as possible. Late Start and Late Finish are calculated based on the project's critical path, helping identify tasks that can be delayed without affecting the project's completion time and those that are critical for the project's timeline. Tasks on the critical path have zero slack or float, meaning any delay in these tasks directly affects the project's timeline.



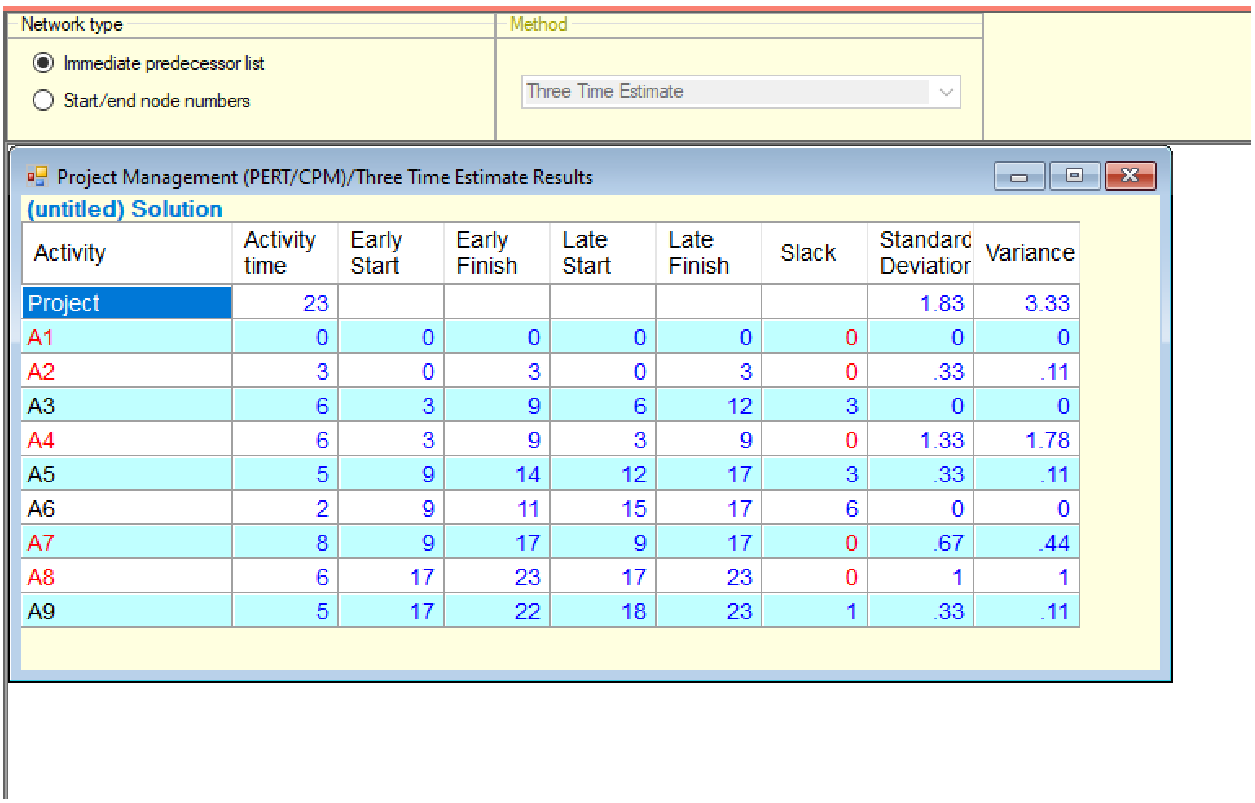
Interpreatation: A Gantt chart is a visual representation of a project schedule that displays tasks or activities along a timeline. While Gantt charts are often used for project planning and tracking, the specific terms like Early Start (ES), Early Finish (EF), Late Start (LS), and Late Finish (LF) are more closely associated with the Critical Path Method (CPM) in project management. The chart displays project tasks along a timeline, with each task represented by a horizontal bar. Associated with CPM terminology, the (ES) is depicted as the initiation point of the task bar, marking the earliest possible start time. Concurrently, the (EF) is illustrated as the termination of the bar, indicating the earliest completion time. (LS) and (LF) values, aids project managers in precisely scheduling activities, identifying the critical path, and ensuring efficient project execution.

Finding the above mention parameters by using PERT Networking Model.

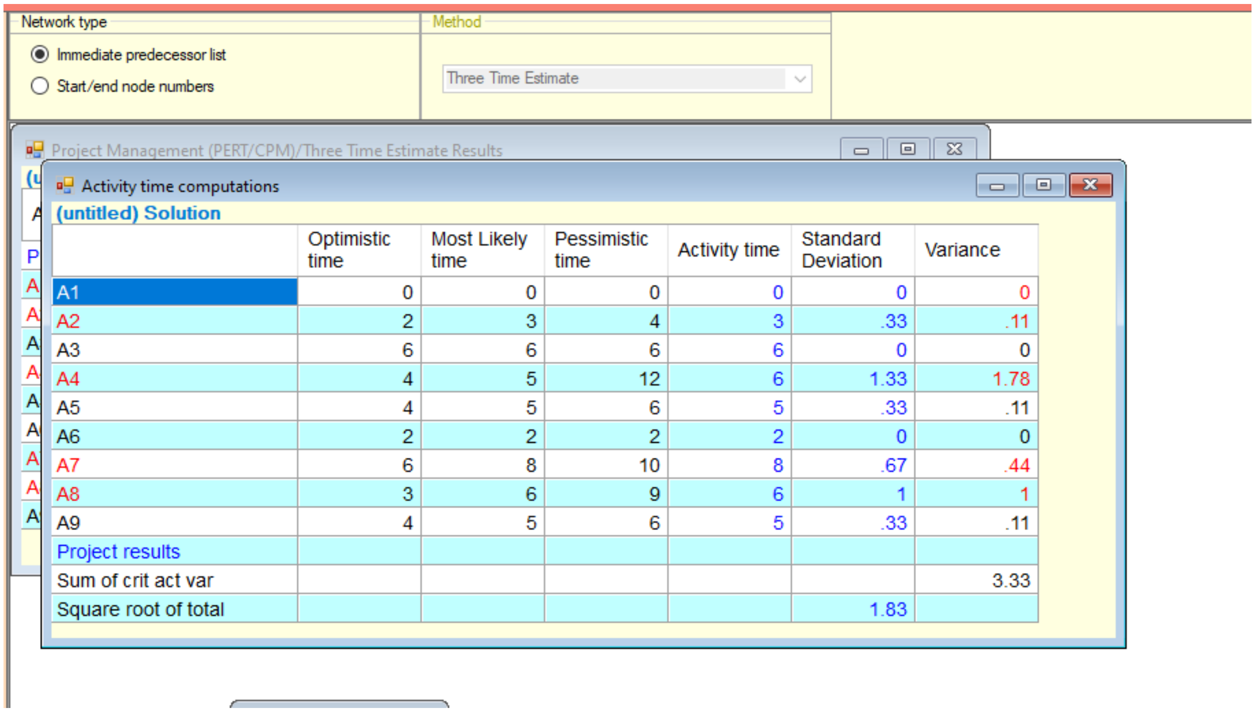
Activity	Optimistic Time (Week)	Most likely Time (Week)	Pessimistic Time (Week)	Immediate Procedure
Agile Eye Mechanism A1	0	0	0	--------
Final CAD Design A2	09	10	17	Agile Eye Mechanism
Base and Support A3	26	26	26	Final CAD Design
Sliding System A4	13	14	15	Final CAD Design, Base and Support
Commanding Motor A5	07	09	11	Final CAD Design, Sliding System
Solar Panel A6	03	06	9	Final CAD Design, Sliding System
Complete Circuit A7	14	15	16	Final CAD Design, Commanding Motor, Solar Panel
Button Parsing A8	06	06	6	Final CAD Design, Commanding Motor, Complete Circuit
Sensor Parsing A9	13	14	15	Final CAD Design, Commanding Motor, Complete Circuit



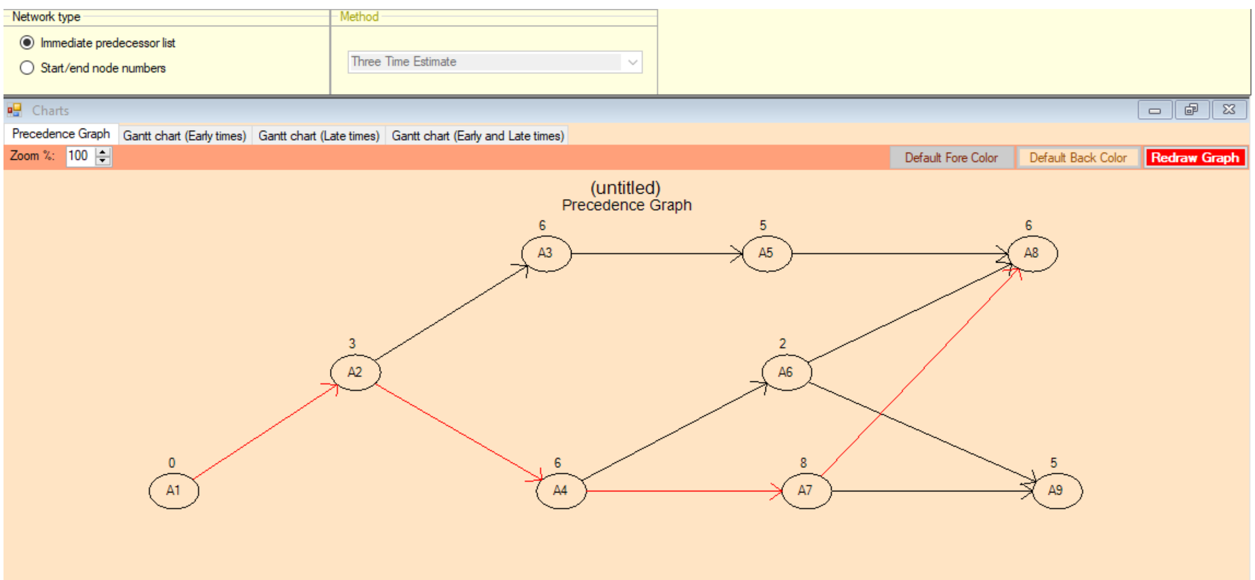
Activity	Optimistic time	Most Likely time	Pessimistic time	Predecessor 1	Predecessor 2	Predecessor 3	Predecessor 4	Predecessor 5	Predecessor 6	Predecessor 7
A1	0	0	0							
A2	2	3	4	A1						
A3	6	6	6	A2						
A4	4	5	12	A2						
A5	4	5	6	A3						
A6	2	2	2	A4						
A7	6	8	10	A4						
A8	3	6	9	A5	A6	A7				
A9	4	5	6	A6	A7					



Activity	Activity time	Early Start	Early Finish	Late Start	Late Finish	Slack	Standard Deviation	Variance
Project	23						1.83	3.33
A1	0	0	0	0	0	0	0	0
A2	3	0	3	0	3	0	.33	.11
A3	6	3	9	6	12	3	0	0
A4	6	3	9	3	9	0	1.33	1.78
A5	5	9	14	12	17	3	.33	.11
A6	2	9	11	15	17	6	0	0
A7	8	9	17	9	17	0	.67	.44
A8	6	17	23	17	23	0	1	1
A9	5	17	22	18	23	1	.33	.11



	Optimistic time	Most Likely time	Pessimistic time	Activity time	Standard Deviation	Variance
A1	0	0	0	0	0	0
A2	2	3	4	3	.33	.11
A3	6	6	6	6	0	0
A4	4	5	12	6	1.33	1.78
A5	4	5	6	5	.33	.11
A6	2	2	2	2	0	0
A7	6	8	10	8	.67	.44
A8	3	6	9	6	1	1
A9	4	5	6	5	.33	.11
Project results						
Sum of crit act var						3.33
Square root of total					1.83	



Interpretation: In (POM) software employing the PERT method, a precedence graph visually depicts task dependencies and their scheduling. Each task is assigned optimistic, most likely, and pessimistic time estimates to account for uncertainties. This approach enhances project management by providing a more realistic and probabilistic view of the project timeline and std deviation and Variance is calculated for the optimistic and pessimistic time represent the uncertainty.

Crashing: We are now going to create a project network diagram and identify the critical path. We will also explore project crashing to determine the project completion time and the associated total cost for completing the project within the normal timeframe. Additionally, we will assess the impact on the total project cost if the project timeline needs to be expedited.

Activity	Normal Time (Week)	Crash Time (Week)	Normal Cost	Crash Cost	Immediate Procedure
Agile Eye Mechanism	0	0	0	0	--------
Final CAD Design	10	09	3000	3750	Agile Eye Mechanism
Base and Support	26	26	5000	6250	Final CAD Design
Sliding System	14	13	8000	10000	Final CAD Design, Base and Support
Commanding Motor	10	07	7000	8500	Final CAD Design, Sliding System
Solar Panel	08	03	10000	15000	Final CAD Design, Sliding System
Complete Circuit	17	14	9000	11000	Final CAD Design, Commanding Motor, Solar Panel
Button Parsing	07	06	5000	7000	Final CAD Design, Commanding Motor, Complete Circuit
Sensor Parsing	16	13	8000	12000	Final CAD Design, Commanding Motor, Complete Circuit

Network type
- (•) Immediate predecessor list
- ( ) Start/end node numbers

Method: Crashing

(untitled)

Activity	Normal time	Crash time	Normal Cost	Crash Cost	Predecessor 1	Predecessor 2	Predecessor 3	Predecessor 4	Predecessor 5	Predecessor 6	Predecess 7
A1	0	0	0	0							
A2	3	2	3000	3750	A1						
A3	6	6	5000	6250	A2						
A4	5	4	8000	10000	A2						
A5	5	4	7000	8500	A3						
A6	2	2	10000	15000	A4						
A7	8	6	9000	11000	A4						
A8	6	3	5000	7000	A5	A6	A7				
A9	5	4	8000	12000	A6	A7					

Network type

- (•) Immediate predecessor list
- ( ) Start/end node numbers

Method: Crashing

Project Management (PERT/CPM)/Crashing Results

(untitled) Solution

Activity	Normal time	Crash time	Normal Cost	Crash Cost	Crash cost/pd	Crash by	Crashing cost
Project	22	16					
A1	0	0	$0	$0	$0	0	$0
A2	3	2	$3000	$3750	$750	1	$750
A3	6	6	$5000	$6250	$0	0	$0
A4	5	4	$8000	$10000	$2000	1	$2000
A5	5	4	$7000	$8500	$1500	0	$0
A6	2	2	$10000	$15000	$0	0	$0
A7	8	6	$9000	$11000	$1000	2	$2000
A8	6	3	$5000	$7000	$666.67	3	$2000
A9	5	4	$8000	$12000	$4000	1	$4000
TOTALS			$55000				$10750

Crash Schedule

(untitled) Solution

Project time	Period cost	Cumulative cost	Activities to crash
22	0	0	
21	666.67	666.67	A8
20	750	1416.67	A2
19	1000	2416.67	A7
18	1000	3416.67	A7
17	2666.67	6083.33	A4, A8
16	4666.67	10750	A8, A9

- (•) Immediate predecessor list
- ( ) Start/end node numbers

Crashing

(untitled) Solution

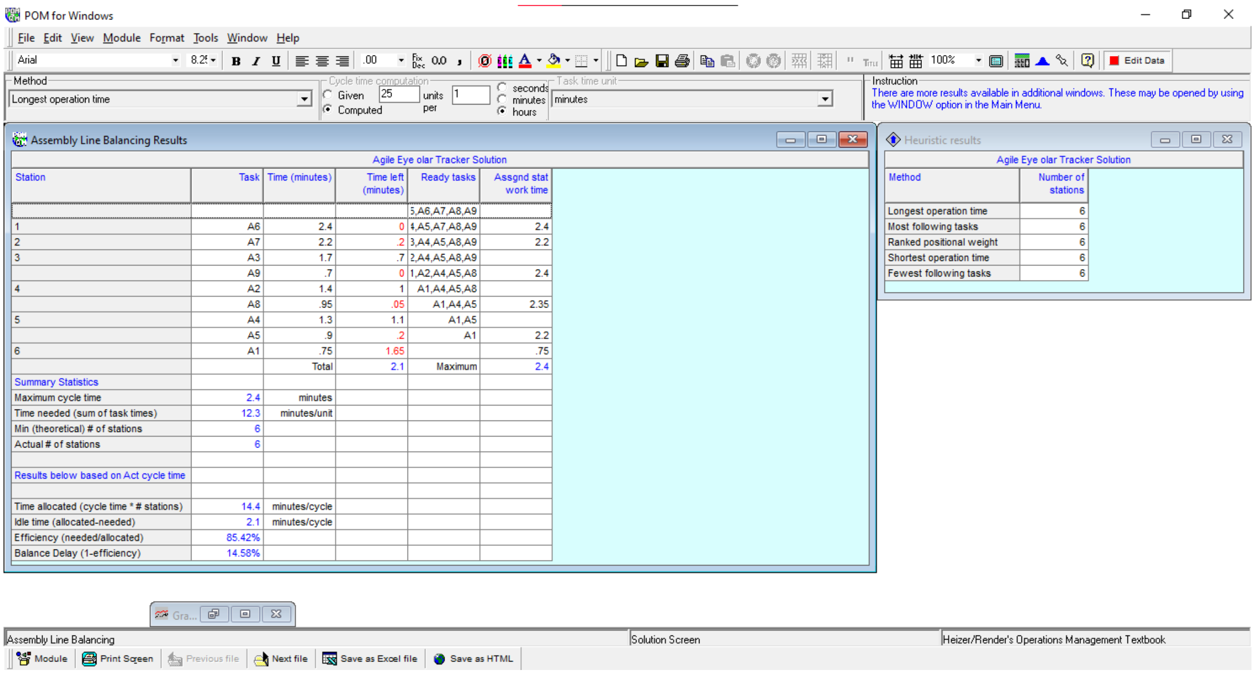
Project time	Period cost	Cumulative cost	A1	A2	A3	A4	A5	A6	A7	A8	A9
22	0	0									
21	666.67	666.67								1	
20	750	1416.67		1						1	
19	1000	2416.67		1					1	1	
18	1000	3416.67		1					2	1	
17	2666.67	6083.33		1		1			2	2	
16	4666.67	10750		1		1			2	3	

Interpretation: In project management, the concept of "crashing" involves expediting project schedules by shortening the duration of critical activities through the allocation of additional resources. When optimizing crashing decisions, project managers consider the normal time and crash time (minimum possible duration with additional resources) of each activity, along with associated normal and crash costs. The decision-making process involves evaluating the trade-off between time and cost, prioritizing critical path activities, and identifying the most efficient way to reduce the project duration within acceptable cost constraints. The outcome is an optimized crashing strategy that minimizes overall project duration while effectively managing costs, aiming to strike a balance between timely project completion and resource efficiency.

Assembly line: Develop a visual representation illustrating the sequence and dependencies of work elements for assembling the Agile Eye Solar Tracker. The production rate is given as 25 units/hour. Calculating the time needed to complete one unit (cycle time) and determining the minimum workforce required to meet the specified production rate. Additionally, identify the maximum time each work element can take (allowable service time) without affecting the overall assembly line balance. These analyses will help optimize the efficiency of the production process.

Work Element Description	T_{ek} (Min)	Preceded By.
Agile Eye Mechanism	0.75	--------
Final CAD Design	1.4	Agile Eye Mechanism
Base and Support	1.7	Final CAD Design
Sliding System	1.3	Final CAD Design
Commanding Motor	0.9	Base and Support
Solar Panel	2.4	Sliding System
Complete Circuit	2.2	Sliding System
Button Parsing	0.95	Commanding Motor, Solar Panel, Complete Circuit
Sensor Parsing	0.7	Solar Panel, Complete Circuit

Longest Operation Time:

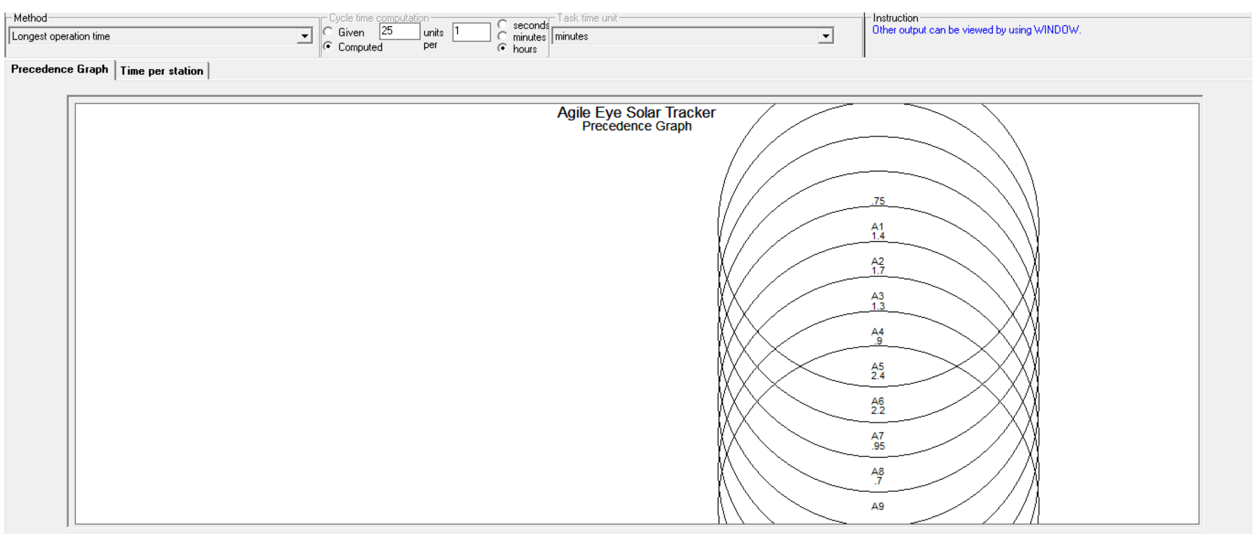


Agile Eye olar Tracker Solution

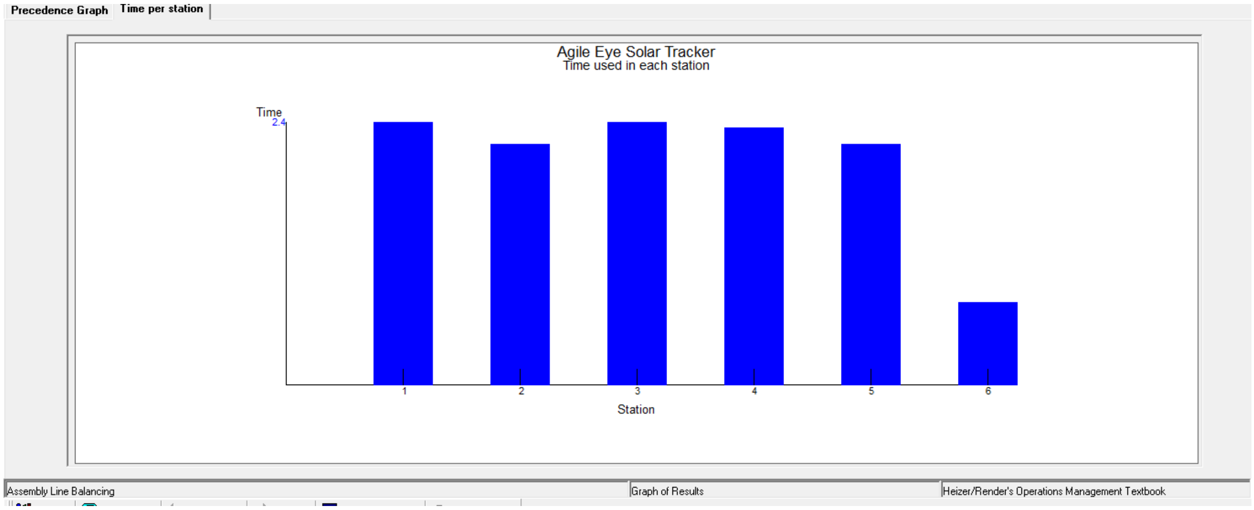
Station	Task	Time (minutes)	Time left (minutes)	Ready tasks	Assgnd stat work time
				5,A6,A7,A8,A9	
1	A6	2.4	0	4,A5,A7,A8,A9	2.4
2	A7	2.2	.2	3,A4,A5,A8,A9	2.2
3	A3	1.7	.7	2,A4,A5,A8,A9	
	A9	.7	0	1,A2,A4,A5,A8	2.4
4	A2	1.4	1	A1,A4,A5,A8	
	A8	.95	.05	A1,A4,A5	2.35
5	A4	1.3	1.1	A1,A5	
	A5	.9	.2	A1	2.2
6	A1	.75	1.65		.75
		Total	2.1	Maximum	2.4
Summary Statistics					
Maximum cycle time	2.4	minutes			
Time needed (sum of task times)	12.3	minutes/unit			
Min (theoretical) # of stations	6				
Actual # of stations	6				
Results below based on Act cycle time					
Time allocated (cycle time * # stations)	14.4	minutes/cycle			
Idle time (allocated-needed)	2.1	minutes/cycle			
Efficiency (needed/allocated)	85.42%				
Balance Delay (1-efficiency)	14.58%				

Agile Eye olar Tracker Solution

Method	Number of stations
Longest operation time	6
Most following tasks	6
Ranked positional weight	6
Shortest operation time	6
Fewest following tasks	6



Interpretation: The longest operation method in assembly line, applied through POM for Windows software, is a strategic approach to optimize production efficiency. It in conjunction with Tek time calculations, this method aims to streamline the assembly line, improving overall efficiency, reducing delays, and enhancing cycle time. Overall, the longest operation method, integrated with Tek time and supported by POM for Windows software, provides a comprehensive framework for achieving operational excellence and aligning production rates with customer demand efficiently.



This graph illustrates each activity work station in the assembly line of Agile eye solar tracker with the particular time.

Ranked Positional Weight:

Method: Ranked positional weight
Cycle time computation: Given / Computed (selected); 25 units per 1; seconds / minutes / hours (selected)
Task time unit: minutes
Instruction: Other output can be viewed by using WINDOW.

Assembly Line Balancing Results

Agile Eye Solar Tracker Solution

Station	Task	Time (minutes)	Time left (minutes)	Ready tasks (positional wt)	Assgnd stat work time
				A8(.95),A9(.7)	
1	A6	2.4	0	A8(.95),A9(.7)	2.4
2	A7	2.2	.2	A8(.95),A9(.7)	2.2
3	A3	1.7	.7	A8(.95),A9(.7)	
	A9	.7	0	A5(.9),A8(.95)	2.4
4	A2	1.4	1	A5(.9),A8(.95)	
	A8	.95	.05	A4(1.3),A5(.9)	2.35
5	A4	1.3	1.1	A1(.75),A5(.9)	
	A5	.9	.2	A1(.75)	2.2
6	A1	.75	1.65		.75
		Total	2.1	Maximum	2.4
Summary Statistics					
Maximum cycle time	2.4	minutes			
Time needed (sum of task times)	12.3	minutes/unit			
Min (theoretical) # of stations	6				
Actual # of stations	6				
Results below based on Act cycle time					
Time allocated (cycle time * # stations)	14.4	minutes/cycle			
Idle time (allocated-needed)	2.1	minutes/cycle			
Efficiency (needed/allocated)	85.42%				
Balance Delay (1-efficiency)	14.58%				

Heuristic results

Agile Eye Solar Tracker Solution

Method	Number of stations
Longest operation time	6
Most following tasks	6
Ranked positional weight	6
Shortest operation time	6
Fewest following tasks	6

Interpretation: The ranked positional weight method in assembly line, executed through POM for Windows software, this method involves assigning weights to tasks based on their positions within the assembly line, prioritizing critical tasks that contribute significantly to operational flow by ranking and prioritizing tasks, the method minimizes delays, optimizes cycle time, and ensures effective resource utilization. The method's consideration of ready tasks and remaining time further enhances its effectiveness in streamlining operations and achieving production targets. Overall, the ranked positional weight method provides a comprehensive framework for efficient task management, contributing to enhanced assembly line performance and alignment with operational objectives.

Shortest Operation Time:

Method: Shortest operation time

Cycle time computation: ( ) Given (•) Computed 25 units per 1 ( ) seconds ( ) minutes (•) hours

Task time unit: minutes

Instruction: There are more results available in additional windows. These may be opened by using the WINDOW option in the Main Menu.

Assembly Line Balancing Results

Agile Eye Solar Tracker Solution

Station	Task	Time (minutes)	Time left (minutes)	Ready tasks	Assgnd stat work time
				5,A6,A7,A8,A9	
1	A9	.7	1.7	4,A5,A6,A7,A8	
	A1	.75	.95	4,A5,A6,A7,A8	
	A5	.9	.05	3,A4,A6,A7,A8	2.35
2	A8	.95	1.45	2,A3,A4,A6,A7	
	A4	1.3	.15	A2,A3,A6,A7	2.25
3	A2	1.4	1	A3,A6,A7	1.4
4	A3	1.7	.7	A6,A7	1.7
5	A7	2.2	.2	A6	2.2
6	A6	2.4	0		2.4
		Total	2.1	Maximum	2.4
Summary Statistics					
Maximum cycle time	2.4	minutes			
Time needed (sum of task times)	12.3	minutes/unit			
Min (theoretical) # of stations	6				
Actual # of stations	6				
Results below based on Act cycle time					
Time allocated (cycle time * # stations)	14.4	minutes/cycle			
Idle time (allocated-needed)	2.1	minutes/cycle			
Efficiency (needed/allocated)	85.42%				
Balance Delay (1-efficiency)	14.58%				

Heuristic results

Agile Eye Solar Tracker Solution

Method	Number of stations
Longest operation time	6
Most following tasks	6
Ranked positional weight	6
Shortest operation time	6
Fewest following tasks	6

Interpretation: The shortest operation time method in assembly line, implemented through POM for Windows software, strategically prioritizes tasks with the shortest processing times to optimize production efficiency. This approach, integrated with considerations for Tek time, efficiency, delay, and cycle time, aims to align the assembly line's performance. By identifying ready tasks with brief operation times, the method minimizes delays the prioritization of tasks considers both their operational efficiency and the time left in the assembly line schedule, ensuring effective utilization of resources to meet production targets. Overall, the method enhances assembly line performance by focusing on efficient task management and aligning production rates with operational goals.

Most Following Tasks:

Method: Most following tasks

Cycle time computation: ( ) Given (•) Computed 25 units per 1 ( ) seconds ( ) minutes (•) hours

Task time unit: minutes

Instruction: There are more results available in additional windows. These may be opened by using the WINDOW option in the Main Menu.

Assembly Line Balancing Results

Agile Eye Solar Tracker Solution

Station	Task	Time (minutes)	Time left (minutes)	Ready tasks (# followers)	Assgnd stat work time
				0),A8(0),A9(0)	
1	A1	.75	1.65	0),A8(0),A9(0)	
	A2	1.4	.25	0),A8(0),A9(0)	2.15
2	A3	1.7	.7	0),A8(0),A9(0)	
	A9	.7	0	0),A7(0),A8(0)	2.4
3	A4	1.3	1.1	0),A7(0),A8(0)	
	A5	.9	.2	0),A7(0),A8(0)	2.2
4	A6	2.4	0	A7(0),A8(0)	2.4
5	A7	2.2	.2	A8(0)	2.2
6	A8	.95	1.45		.95
		Total	2.1	Maximum	2.4
Summary Statistics					
Maximum cycle time	2.4	minutes			
Time needed (sum of task times)	12.3	minutes/unit			
Min (theoretical) # of stations	6				
Actual # of stations	6				
Results below based on Act cycle time					
Time allocated (cycle time * # stations)	14.4	minutes/cycle			
Idle time (allocated-needed)	2.1	minutes/cycle			
Efficiency (needed/allocated)	85.42%				
Balance Delay (1-efficiency)	14.58%				

Heuristic results

Agile Eye Solar Tracker Solution

Method	Number of stations
Longest operation time	6
Most following tasks	6
Ranked positional weight	6
Shortest operation time	6
Fewest following tasks	6

Interpretation: The most following task method in assembly line management, using POM for Windows software, is a way of organizing tasks by focusing on those that have the most impact on subsequent steps in production. The method aims to make the assembly line more efficient and minimize delays. The approach considers time constraints and aims to streamline operations for improved overall performance of the assembly line.

Fewest Following Task:

Method: Fewest following tasks

Cycle time computation: Given / Computed 25 units per 1 seconds / minutes / hours

Task time unit: minutes

Instruction: There are more results available in additional windows. These may be opened by using the WINDOW option in the Main Menu.

Assembly Line Balancing Results

Agile Eye Solar Tracker Solution

Station	Task	Time (minutes)	Time left (minutes)	Ready tasks (# followers)	Assgnd stat work time
				0),A8(0),A9(0)	
1	A1	.75	1.65	0),A8(0),A9(0)	
	A2	1.4	.25	0),A8(0),A9(0)	2.15
2	A3	1.7	.7	0),A8(0),A9(0)	
	A9	.7	0	0),A7(0),A8(0)	2.4
3	A4	1.3	1.1	0),A7(0),A8(0)	
	A5	.9	.2	0),A7(0),A8(0)	2.2
4	A6	2.4	0	A7(0),A8(0)	2.4
5	A7	2.2	.2	A8(0)	2.2
6	A8	.95	1.45		.95
		Total	2.1	Maximum	2.4
Summary Statistics					
Maximum cycle time	2.4	minutes			
Time needed (sum of task times)	12.3	minutes/unit			
Min (theoretical) # of stations	6				
Actual # of stations	6				
Results below based on Act cycle time					
Time allocated (cycle time * # stations)	14.4	minutes/cycle			
Idle time (allocated-needed)	2.1	minutes/cycle			
Efficiency (needed/allocated)	85.42%				
Balance Delay (1-efficiency)	14.58%				

Heuristic results

Agile Eye Solar Tracker Solution

Method	Number of stations
Longest operation time	6
Most following tasks	6
Ranked positional weight	6
Shortest operation time	6
Fewest following tasks	6

Interpretation: The fewest following task method in assembly line, using POM for Windows software, prioritizes tasks with the least impact on activities. This approach aims to streamline production by giving importance to tasks that have fewer dependencies, minimizing delays and optimizing the overall cycle time. It considers task readiness and the time left in the assembly line schedule, offering a systematic and efficient approach to task management for improved assembly line performance.

"In a comparative analysis, it seems that the efficiency of the assembly line is lower when using the shortest time, ranked positional weight, and fewest following task modules, in contrast to the longest operation method. The longest operation method appears more effective, possibly due to its focus on addressing critical points in the assembly line. The choice between these methods may depend on factors such as task dependencies, resource availability, and overall production goals"

6. Identifications of areas where use of modern tool is required.

In project management, the use of modern tools is essential across various areas, including planning, collaboration, communication, task and resource management, document handling, reporting, risk management, budgeting, and client collaboration. These tools improve efficiency, streamline processes, and contribute to successful project outcomes. Modern tools widely used in industries for project management, Slack and Microsoft Teams for collaboration, Zapier and IFTTT for task and workflow, Google Drive and Dropbox for cloud storage, Todoist and Notion for task tracking and productivity, SAP IBP and Oracle SCM Cloud for supply chain management, these tools cater to various business needs, enhancing efficiency, collaboration, and management across different sectors.

7. Expected Outcomes:

The expected outcomes of the project management scenario include improved execution of the innovative solar tracker design project in the electric vehicle (EV) industry. By utilizing tools like POM-QM for Windows, the project anticipates enhanced accuracy in planning, efficient resource management, cost control, and timely completion. Anticipated benefits encompass enhanced accuracy in project planning, efficient resource management, cost control, and timely project completion through the use of tools like POM-QM for Windows and methodologies such as the Critical Path Method (CPM) and Program Evaluation and Review Technique (PERT). Overall, the project strives for continuous improvement in project management practices and technological advancements within the EV and sustainable energy sector.

8. Conclusion:

On the whole, the project focuses on developing an innovative solar tracker design in the EV industry, aiming to overcome challenges through improved project management practices. Utilizing tools like POM-QM for Windows and methodologies such as CPM and PERT, the project anticipates enhanced planning accuracy, efficient resource management, cost control, and timely completion. With an emphasis on mechatronics and agile mechanisms, the project seeks to deliver an advanced and energy-efficient solution. Overall, the project sets a standard for continuous improvement in project management and technological advancements in the EV industry.

9. Reference links:

https://www.instructables.com/Agile-Eye-Solar-Tracker/

“The project management information has been sourced from this website, while the remaining values and data, such as activity time, cycle time, and Tek time, as well as their costs, were assumed. All calculations and analyses were performed using POM-QM Windows versions 4 and 5 software”